

6. Sensory and motor areas of the cerebral cortex (I)

The whole of the cerebral cortex was divided into 52 different areas in an early presentation by Korbinian Brodmann. These areas known as Brodman areas, are based on their cytoarchitecture but also relate to various functions. An example is Brodmann area 17 which is the primary visual cortex.

In more general terms the cortex is typically described as comprising three parts: sensory, motor, and association areas (figure 28).

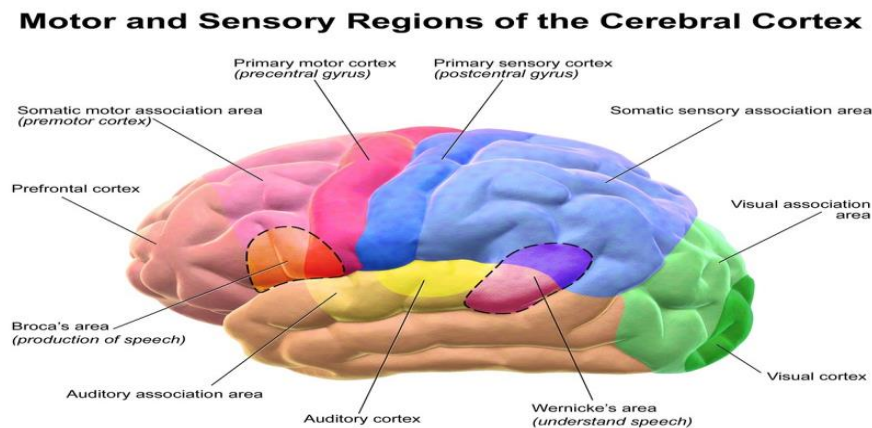


Figure 28 : Areas of the cortex

Sensory areas

- The sensory areas are the cortical areas that receive and process information from the senses.
- Parts of the cortex that receive sensory inputs from the thalamus are called primary sensory areas.
- The senses of vision, hearing, and touch are served by the primary visual cortex, primary auditory cortex and primary somatosensory cortex respectively.
- In general, the two hemispheres receive information from the

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opposite (contralateral) side of the body.

- For example, the right primary somatosensory cortex receives information from the left limbs, and the right visual cortex receives information from the left visual field.

SENSORY CORTEX

- No sensation (somatic or special) is allowed to the cerebral cortex without relay in the thalamus first EXCEPT Olfaction.
- Somatic sensory areas of the cerebral cortex are classified into:

1. Primary somatic sensory areas:

- Somatic sensory area I.
- Somatic sensory area II.

2. Somatic association areas.

1. Primary somatic sensory areas:

- Somatic sensory area I : (areas 3,1,2)
 - Site: In the postcentral gyrus (areas 1,2,3), immediately behind the central sulcus in the parietal lobe.
 - Afferent connections: It receives sensory impulses from PLVNT.
 - It is characterized by (figure 29):
 - 1) The representation is crossed: The somatic sensory area I of one side receives somatic sensations from the opposite side of the body.
 - 2) The representation is inverted: The head is represented downwards, while the trunk and lower

limbs upwards, and the leg on the medial surface.

3) The greater the sensory function of the part (increased number of sensory receptors), the larger is the size of its area of representation.

- Function: It is the center for the following sensations:
 - Fine touch: tactile localization, tactile discrimination, stereognosis and texture of material.
 - Pressure sensation: discrimination of weights.
 - Vibration sense.
 - Kinesthetic sensation: sense of position and of movements.
 - Discrimination of various grades of temperature.

Cortical plasticity:

The cortical area of representation of a given part of the body is not absolutely fixed, but it is liable to variations under different conditions. This is called plasticity of the cortex.

Examples:

- Repeated stimulation of a specific sensory pathway from a given part → widens the cortical area of representation of the part by training.
- When a limb is amputated, its cortical area of representation does not show disuse atrophy, but is associated with that of the surrounding limbs.
- Conversely, when the cortical area of a specific limb is removed, a new cortical area of representation is created in the surrounding cortex so long as the afferent sensory pathways are intact.

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Cortical plasticity depends on:

- Experience and training.
- Wide subcortical connections of different cortical areas.
- Divergence and convergence of afferent sensory pathways on cortical neurons by means of intercommunicating neurons.

2. Somatic sensory area II

- Site: A small area behind the lower part of postcentral gyrus (somatic sensory area I) in the upper wall of the lateral sulcus.
- Afferent connection:
 - It receives sensory impulses from PLVNT.
 - It receives afferent connection from somatic sensory area I serving learning.
- Characters:
 - The face area lies anterior, the arm and trunk areas in the middle and the lower limbs posterior. However, there is no sharp demarcation between the different areas as present in somatic sensory area I.

Functions:

- It is a cortical center for pain sensation.
- It serves sensory control of motor functions.
- It serves learning based on tactile information from sensory area I.
- N.B. Mental processing in somatic sensory area II is dependent on informations relayed to it from sensory area I and not the reverse.

Clinical significance:

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Damage of somatic sensory area of I:

- loss of sensations (position and stereognosis).
- Failure of learning by somatic sensory area

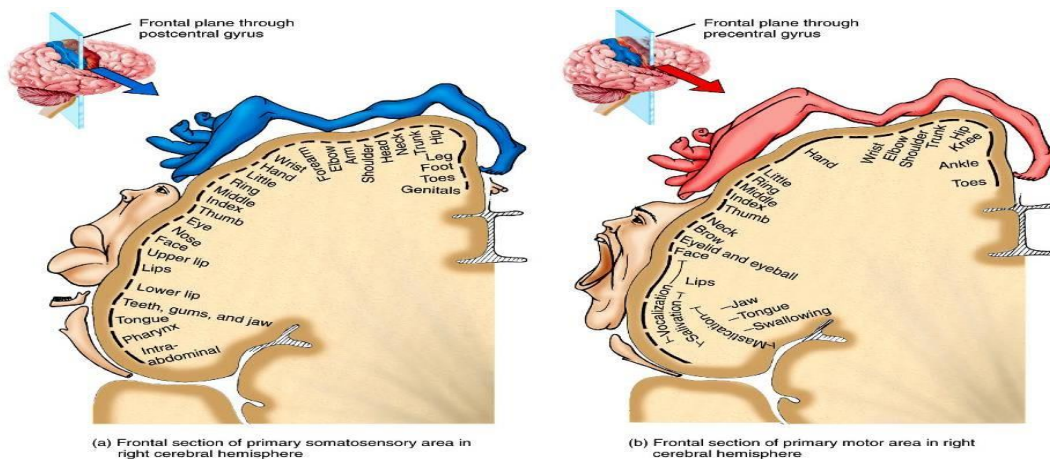
II. Damage of somatic sensory area II :

- Failure of learning based on tactile discrimination.

2. Sensory association area(areas 5, 7)

- Site: In the parietal lobe immediately behind area I and above area II.
- Afferent connections: It receives impulses from: -
 1. Primary sensory areas I & II.
 2. PLVNT.
 3. Non specific thalamic association nuclei.

- Function:



Associate the various types of sensations perceived by the primary sensory areas so that they give a meaning to what we feel. It depends on memory and experience.

Clinical significance:

Damage or removal of the somatic association sensory areas leads to

1. Loss of the ability of the person to recognize complex objects.

2. The person may forget completely the opposite side of the body.

Figure 29: sensory (a) and motor (b) areas of representation of the body

Clinical Syndromes Associated With Sensory Disturbances

1. Syringomyelia

- Cause: Cystic dilatation of the central canal of the spinal cord due to cavitation in the gray matter around it.
- Effect: Destruction of the fibers of the lateral and ventral spinothalamic tracts which cross in front of the site of the lesion.
- Results: Loss of pain, temperature and crude touch sensations on both sides of the body at the levels of the segments affected. Fine touch is not affected as it ascends directly in the posterior column (dissociated sensory loss).
- Common sites:
 - In the cervical enlargement of the spinal cord → jacket distribution of dissociated sensory loss.
 - The lower limbs may be also affected in the lumbar type.

2. Tabes dorsalis:

- Cause: It is due to syphilis (a sexually transmitted disease) .
- Effect: Syphilis produces inflammation of the dorsal root ganglia → compression of the posterior root fibers leading to:
 - At first we get irritation (↑ discharge) followed by degeneration (loss of sensations)
 - Large myelinated fibers of the posterior root (dorsal column) degenerate earlier than thin myelinated and nonmyelinated

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fibers.

Clinically:

- Pain:

- Attacks of severe pain due to irritation of the pain fibers in the dorsal roots.
- Pain may be referred to the skin dermatomes affected.
- Later on, pain fibers degenerate → loss of pain sensation
- The slow component of cutaneous pain (burning pain) remains till a late stage as it is transmitted by C nonmyelinated fibers which resist pressure.

- Sensory ataxia:

- It is due to degeneration of dorsal column fibers (Gracile and Cuneate tracts).
- It is characterized by:
 - Loss of fine touch sensation (tactile localization, tactile discrimination, stereognosis, and texture of material).
 - Loss of vibration sense: When a vibrating tuning fork is placed over a bony prominence, the patient does not feel vibration but only a cold object touching the skin.
 - Loss of kinesthetic and pressure sensation → incoordination of voluntary movements due to loss of sense of position and of movement.
 - Loss of pressure sensation → stamping gait.
 - Positive Romberg's sign: The patient can keep his balance better and coordinate his movements by using his eyes, but if he closes

his eyes or he is put in the dark, he sways and may even fall.

- Loss of all superficial and deep reflexes.

Common site: The roots of the lumbosacral and brachial plexuses are commonly affected. The common areas of loss of sensation are around the anus, lower limbs, upper chest and ulnar border of hands.

3. Herpes Zoster

- Viral disease caused by Herpes virus (associated with chicken pox) which infects and irritates the cell bodies of the dorsal root ganglia (DRG) → severe pain in the thoracic. dermatomes affected, commonly the upper thoracic.
- Collateral branches from the irritated posterior roots end on autonomic ganglia → stimulate autonomic reflex arcs → * cutaneous vasodilatation (redness) and increased capillary permeability forming cutaneous vesicles. • The condition is usually unilateral.

4. Spinal cord lesions:

- They are common clinical cases frequently met with and may be in the form of:
 - Acute injuries: e.g. gun shot injuries, fractures of the vertebral column vascular trauma and inflammation.
 - Chronic injuries: e.g. compression of the cord by tumors from within or from outside the cord.
- They may result in:
 - Complete transection: where all ascending and descending tracts are cut.
 - Incomplete (partial) transaction: where only some tracts are cut and the others are not.

1. Hemisection of the cord:

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The sensory disorders associated with this disease are as follows:

A. Above the level of the lesion:

Irritation of dorsal roots just above the lesion corresponding skin dermatomes.

B. At the level of the lesion:

Loss of all sensations from the skin dermatomes supplied by the posterior roots that enter the damaged segments On The Same Side Of The Lesion.

C. Below the level of the lesion:

- Damage of dorsal column → loss of fine touch, vibration, pressure and kinesthetic sensation On The Same Side Of The Lesion,
- Damage of Spinothalamic tracts (lateral and ventral) → loss of pain, temperature and crude touch sensation On The Opposite Side Of The Lesion.

2. Complete transection of the cord:

According to the site of section, the following effects are produced:

- a) At the upper cervical segments (Above the origin the phrenic nerve, above C₃) → failure of respiration: immediate death as in hanging.
- b) At the lower cervical segments (below the origin the phrenic nerve, above C₅) → respiration is normal but there is paralysis of all four limbs (Quadriplegia).
- c) In the thoracic region (below T₂ , i.e. below the origin of the brachial plexus) respiration and the upper limbs are normal but there is paralysis of the lower limbs (paraplegia).

In both quadriplegia and paraplegia, the patient's life is saved but the patient passes

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into the following stages:

I- The stage of spinal shock: This stage occurs immediately after transection of the spinal cord and the manifestations are due to:

- Failure of the spinal cord centers below the level of transection when they become separated from the higher centers due to cutting the descending tracts.
- Section of both the ascending (sensory) and descending (motor) tracts → sensory and motor disturbances below the level of the lesion.
- The manifestations are:
 - A-Cutting the ascending tracts leads to:
 - There is permanent loss of all sensations from all parts of the body supplied by the dorsal roots that enter the spinal cord below the level of the lesion (level sensory loss).
 - Temporary pain may be present in the early days due to:

Irritation of the dorsal roots at the site of the lesion → corresponding dermatomal pain and hyperalgesia.

Irritation of the central cut end of the lateral spinothalamic tracts → pain sensation projected to the affected parts of the body below the lesion.

Pain disappears soon after few days due to degeneration of the cut fibers of the posterior roots and spinothalamic tracts.

B-Cutting the descending motor tracts from the cerebral cortex to AHCS leads to:

- There is permanent loss of all voluntary movements (paralysis) performed by muscles supplied by AHCS below the level of the lesion.

* Lower cervical lesions → quadriplegia.

* Thoracic lesions below T₂ → paraplegia.

C-Cutting the descending tracts from the vasomotor centers in the medulla to LHCS in the thoraco-lumbar region leads to: -

- There is temporary loss of LHCS' discharge below the level of the lesion → temporary loss of sympathetic vasoconstrictor tone → vasodilatation and drop of blood pressure.
- The lower is the level of the lesion, the lesser will be the drop in pressure.
- A lesion below L₃ → no drop in blood pressure.

D-Spinal cord reflexes:

- All spinal cord reflexes whose spinal centers lie below the level of the lesion are lost during this stage as follows:
 - Loss of flexion withdrawal reflex → ulceration (bed sores) of the pressure points of the body due to prolonged recumbence.
 - Loss of the two components of the stretch reflex: -
 - Loss of the static component → loss of muscle tone → flaccid paralysis.
 - Loss of the dynamic component → loss of tendon jerk.
- Significance: Loss of muscle tone → ↓ venous return → ↓ cardiac output → ↓ blood flow to the limbs → The limbs become cold, cyanotic (due to stagnant hypoxia), and dry (due to absent sweat secretion) → predispose to ulceration.
- Changes in normal micturition reflex:

Loss of the normal micturition reflex → retention with overflow Mechanism:

- Due to sensory loss → permanent loss of conscious desire of micturition.

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- Due to voluntary motor loss → paralysis of the external urethral sphincter
 - - Striated muscle) → permanent loss of voluntary motor control on the act of micturition.
- Due to temporary paralysis of the sacral (parasympathetic, S_{2,3,4}) centers → the bladder wall is paralyzed (acts as elastic reservoir) and the internal urethral sphincter shows increased tone.
- The micturition reflex changes into retention with overflow i.e. the bladder continues to fill until the intravesical pressure exceeds the tone (resistance) of the internal urethral sphincter → dribbling of extra urine added

N.B.

- The bladder never empties its content completely.
- Stagnation of urine in the bladder predisposes to infection and stone precipitation.

- Changes in normal defecation reflex:

Like the changes in micturition reflex. Loss of the normal defecation reflex → retention with overflow. The colon is paralyzed and distended with feces and the external anal sphincter is paralyzed.

- Loss of erection and ejaculation reflexes.

E- Lesions in the lower cervical or upper thoracic segments destruction of the cervical sympathetic centers → interruption of the pupillary dilator fibers → constriction of the pupil (Horner's syndrome).

Causes of Spinal shock:

- 1- Spinal shock is not due to decreased blood pressure because: -
 - a. It occurs in transverse section of spinal cord in the lower lumbar region where there is no drop of blood pressure.
 - b. In transverse section of spinal cord in the thoracic region there is marked

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drop of blood pressure, however, the spinal reflexes whose centers lie above the section never show spinal shock.

- 2- Spinal shock is not due to trauma of spinal cord (traumatic shock) because: -
 - a. It occurs equally in clean cut injury (minimal tissue damage) as well as by massive trauma (severe tissue damage) of spinal cord.
 - b. If after recovery from spinal shock another section is made below the first, it does not produce a new state of shock in spite of the second tissue damage.
- 3- Spinal shock is due to cutting the descending facilitatory tracts to spinal cord reflex centers → enter into a state of central inhibition Evidence:

Evidence:

Facilitatory tracts descend in the Ventral quadrant while inhibitory tracts descend the dorsolateral quadrant of spinal cord so, incomplete section of the cord affecting the dorsolateral quadrant →

- loss of inhibitory tracts,
- unantagonized facilitation, and
- exaggerated reflexes (hypertonia and hyper-reflexia)

Duration of Spinal shock:

It continues for sometimes depending on the degree of development of the animal in the animal kingdom (brain development). It is few minutes in the frog, few hours in cats or dogs, few weeks in the monkey and human (2-6 w).

II- The stage of recovery:

The central inhibitory state (that occurred during the shock stage) of the spinal cord centers below the section gradually disappears and the centers restore their excitability and start to respond to afferent stimulation independent of the higher centers.

- The manifestations are:

A- There is no regeneration of tracts inside the CNS so, loss of sensations is

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permanent.

- B- Recovery of blood pressure: The lateral horn cells of the sympathetic thoracolumbar vasoconstrictor spinal cord centers start to discharge independent of higher VMC vasoconstriction gradual ↑ blood pressure to normal.
- C- Somatic spinal cord reflexes recover gradually in the following order:
 - The flexion withdrawal reflex appears earlier than the extensor reflexes.
 - Superficial skin reflexes:
 - i. The plantar reflex shows positive Babinski sign: Stroking the skin of the sole of the foot → dorsiflexion of the big toe (pyramidal tract lesion + dorsiflexion and fanning of the lateral four toes (extrapyramidal tract lesion).
 - ii. It is a type of flexion withdrawal reflex.
 - iii. When fully developed, it is associated with dorsiflexion at the ankle, flexion at the knee and hip.
 - iv. It is accompanied with reciprocal inhibition of the extensors.
 - v. It is normally inhibited by the pyramidal and extrapyramidal tracts to get the normal plantar response.
 - vi. Positive Babinski sign is also present in:
 - 1. Normal children below one year due to incomplete myelination of pyramidal tracts (still functionless).
 - 2. Adults with depressed function of cerebral cortex e.g. during : sleep, coma, and anaesthesia.
 - 3. Upper motor neurone lesions.
 - The deep reflexes:
 - The Stretch reflex:
 - The dynamic component (tendon jerk) appears earlier than the static component (muscle tone).

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- The jerks are slightly exaggerated but with no clonus due to hypotonia.
- The tone reappears (but weak) earlier in the flexors than the extensors → flexor attitude of the paralyzed limbs → paraplegia in flexion
- The skin circulation of the paralyzed limbs improves due to:
 - ↑ B.P, - Recovery of muscle tone, and Recovery of protective reflexes leading to:
 - ↑ skin temperature,
 - disappearance of cyanosis,
 - prevents muscle wasting,
 - helps recovery of trophic ulcers.
- The autonomic reflexes:

A) Reflex micturition shows Automatic Micturition

(Bladder): Characters:

- Sensory loss is permanent (never recovers) → permanent conscious desire of micturition.
- Voluntary motor loss is permanent (never recovers) loss of voluntary motor control on the act of micturition.
- The micturition reflex changes into automatic micturition (automatic bladder) i.e. the bladder fills and the increased intravesical pressure stimulates the sacral autonomic centers to produce contraction of the bladder wall and relaxation of the internal urethral sphincter bladder → emptying helped by:
 - recovery of the sacral (parasympathetic, S_{2,3,4}) centers → start discharge.
 - recovery of tone of the bladder wall.
 - the increased tone of the internal sphincter during spinal shock

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recovers back to normal.

- N.B. = The bladder is efficiently emptied by this reflex. = If the abdominal muscles are not paralyzed, the patient may voluntarily empty the bladder by contracting his abdominal muscles → ↑ intra-pressure → bladder evacuation.

B) Reflex defecation show Automatic defecation (Automatic Rectum). The same as automatic bladder.

C) The Coitus Reflex: * Stimulation of the glans penis → produce the following reflexes: - reflex erection and ejaculation, - contraction of the cremaster muscle with elevation of the testicles, - contraction of dartos muscle with curling of scrotal skin, and - contraction of recti abdominis, adductors of the thigh and flexors of the hip. * Sexual orgasm never occurs.

D) The Mass Reflex: * Scratching the skin below the level of the lesion leads to: - micturition and defecation, - profuse sweating below the level of the lesion. - swinging of blood pressure (unstable ↑ and ↓), and - flexor spasms of the lower limb and abdominal muscles. It is a flexion withdrawal reflex where the afferent impulses irradiate to the facilitated lower spinal cord centers below the lesion. Cause of facilitation:

Advanced recovery occurs due to:

Good nursing care of:

- the skin to prevent bed sores by frequent changing position in bed and cleaning with alcohol.
- the bladder by catheterization and proper antibiotics to prevent retention and infection..
- Maximal recovery occurs nearly after 6 months with the following:
 1. The tone in the extensors becomes greater than in the flexors so the limbs become extended paraplegic → Paraplegia in extension

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manifestations:.

2. The flexion withdrawal reflex becomes associated with strong and apparent crossed extensor reflex.
3. The positive supporting reflex becomes well developed and may permit standing for sometimes unsupported.
4. The spinal stepping reflexes appear and passive flexion of one limb → crossed extension of the opposite limb.
5. The mass reflex

disappears. The stage of failure of recovery: -

- It occurs if proper medical care is neglected → bed sores + urine retention + → toxins → inhibit the spinal cord centers →
- Automatic bladder fails retention with overflow.
- Muscle tone and tendon jerks become weak, then disappear as in spinal shock.
- The patient usually dies from urinary tract infection and renal failure.

3. Quadrant lesions:

1) Posterior Quadrant Lesion:

- Due to cutting the gracile and cuneate tracts, there is loss of proprioceptive sensation on the same side below the lesion.
- Due to cutting the descending dorsoventral inhibitory tracts, there is spastic paralysis on the same side below the lesion.

2) Anterior Quadrant Lesion:

- Due to cutting the lateral and ventral spinothalamic tracts, there is loss of pain, temperature and crude touch sensations on the opposite side below the lesion.
- Due to cutting the descending anterior facilitatory tracts, there is

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hypotonia and flaccidity on the same side below the lesion. There is no paralysis.

5. Thalamic syndrome:

Cause : Vascular lesion (thrombus, embolus, hemorrhage) of the thalamogeniculate artery which supplies PLVNT. The neurones of the PLVNT degenerate, while the anterior and medial thalamic nuclei remain intact.

Clinically: -

- Loss of all sensations from the opposite side of the body.
- Loss of kinesthetic sensations → sensory ataxia on (incoordinated voluntary movements i.e. ataxia).
- After few months, crude sensations (protopathic) return in the affected side characterized by:
 1. They are poorly localized.
 2. They are of high threshold (need strong stimulation).
 3. They are painful regardless of the stimulus applied to the body the opposite side.
 4. They are very unpleasant and accompanied by severe emotional reactions.

Mechanism:

- The medial thalamic nuclei become facilitated and increase the sensitivity to pain transmission through the reticular formation as well as to limbic system Mechanism: leading to emotional disturbance.
- Muscle weakness.
- Epicretic sensations: are the fine sensations perceived by the cerebral cortex.
- Protopathic sensations: are crude sensations perceived by the thalamus.